

GCI WORLD 2026 • FINAL ASSIGNMENT

Stop the Leak: A Data-Driven Retention Engine for Company A

Turning 100k customers of usage history into a targeted, ROI-positive churn-prevention program.

Executive summary

Company A loses a large share of its base each cycle. We built a churn-risk model on its own usage data, identified the customers most likely to leave, and show that a focused retention offer to the riskiest 10% pays for itself several times over.

0.695

Best model AUC (XGBoost)

1.6x

more churners found vs random targeting

\$1.2M

net benefit / year (illustrative)

AGENDA

1. Market context
2. The data & EDA
3. Problem definition
4. Modelling & evidence
5. Risk-based targeting
6. Business proposal
7. Quantified impact
8. Pricing & next steps

A saturated market where retention beats acquisition

21.5%

global telecom churn rate, 2025¹

5–7×

costlier to acquire than to retain a customer²

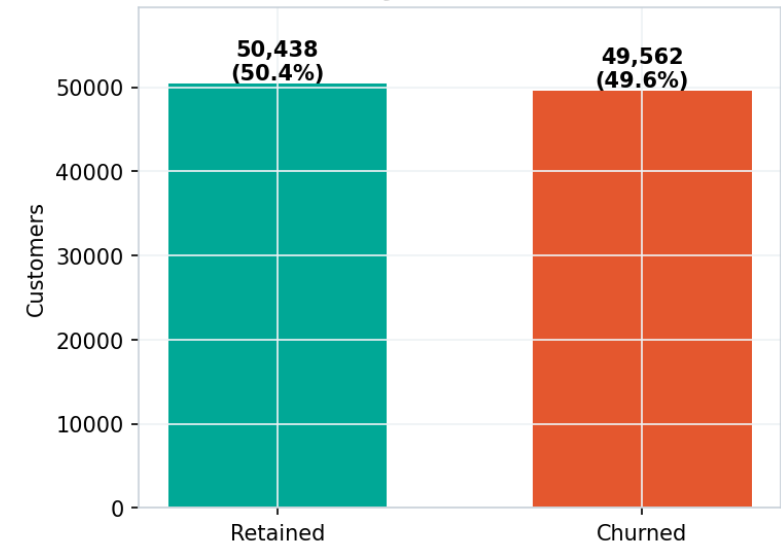
\$120M

lost yearly at 1M subs · \$50 ARPU · 20% churn³

Why this is the right problem to solve

- Mature markets grow by keeping customers and lifting revenue per user, not by net adds.
- Behavioural signals — usage decline, complaints, ageing handsets — precede churn by months, so it is predictable.
- A retained customer is far cheaper than a replacement, so even a modest save rate is highly profitable.

Customer churn is a major, addressable share of the base



The asset: Company A's own customer data

100,000 customers • 100 fields

Two linked tables joined on Customer_ID:

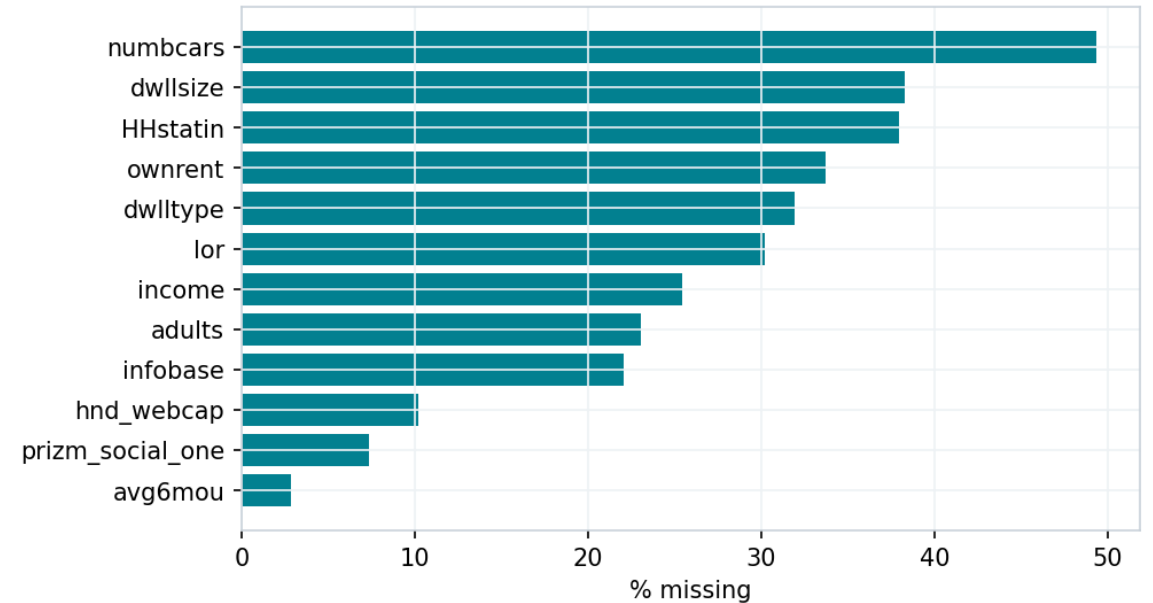
- Client — tenure, plan, equipment, demographics
- Record — monthly usage, billing, call-quality + the churn flag

churn = customer left 31–60 days after the observation date.

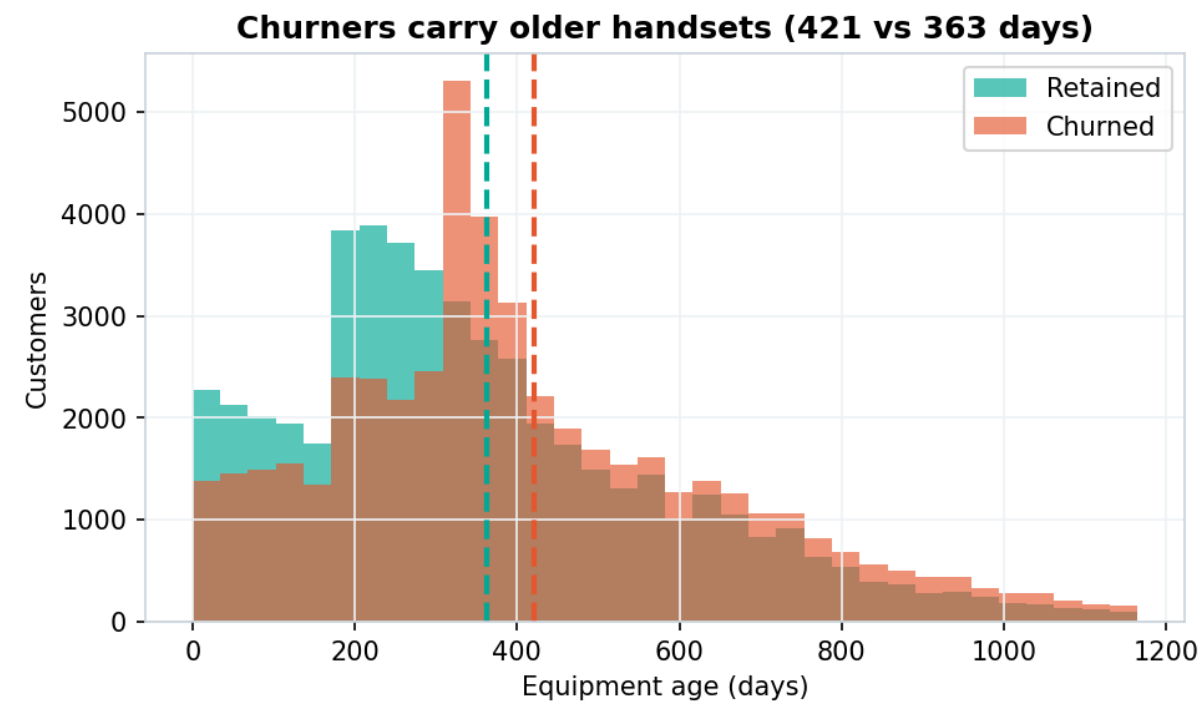
First task: choose the question

The data supports many targets. We chose churn because it is directly tied to revenue, the signals to predict it are present, and — crucially — the output is actionable: a retention team can act on a ranked risk list tomorrow.

Data-quality scan: most-incomplete fields



EDA finding #1: ageing handsets are the clearest churn lever



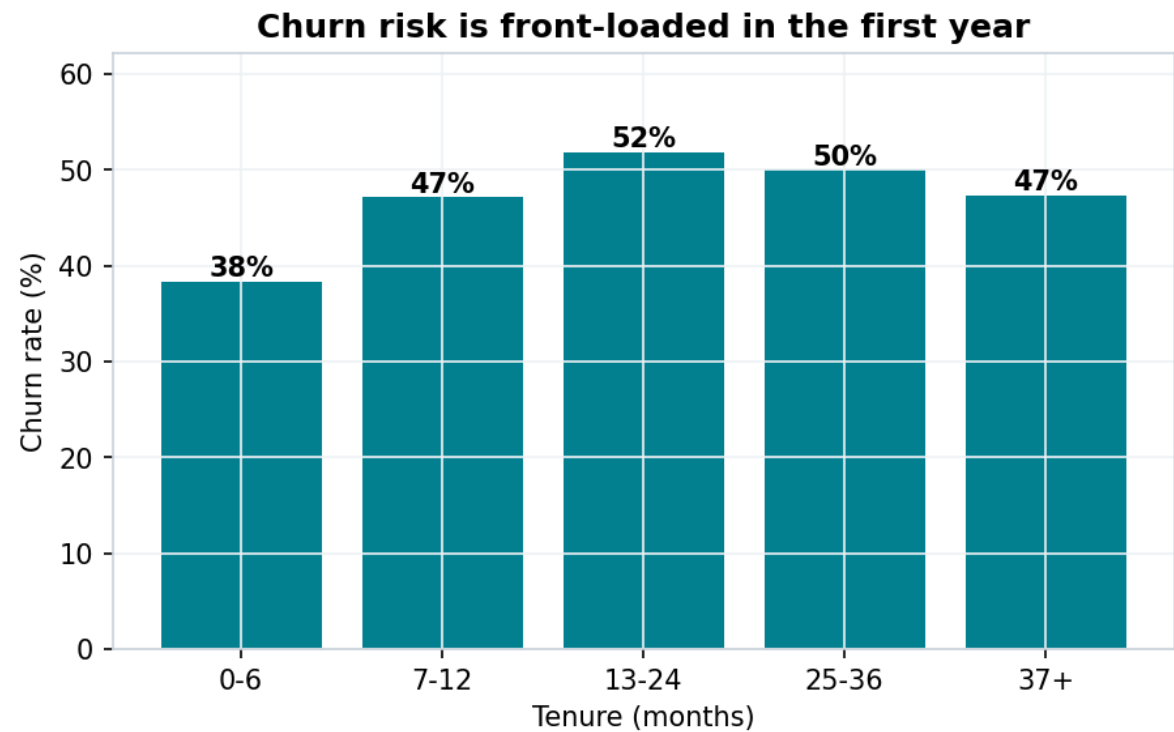
Read this chart

Churners carry markedly older equipment on average (421 days vs 363 for those who stay).

Why it matters for a proposal

Unlike tenure or demographics, equipment age is something the business can change — through upgrade offers. That makes it a lever, not just a correlate.

EDA finding #2: risk is concentrated early in the lifecycle



Tenure bands in months.

Where to focus

Churn rate is highest in the first 12 months and falls as customers settle in. New, under-tenured customers with old handsets are the highest-priority segment.

This shapes the proposal

A retention program should weight early-tenure, declining-usage customers — exactly the group the model will surface in its top risk decile.

From a vague worry to a precise, answerable question

The business question

“Which current customers are most likely to leave in the next cycle, and is it worth intervening?”

ML task

Binary classification. Target = churn. Output = a 0–1 risk score per customer.

Who acts on the output

The retention team receives a ranked list. High-risk, high-value customers get a targeted offer before they leave.

Why a score, not a label

A ranked score lets the business choose how many customers to contact based on budget — a single yes/no label cannot.

Engineering signals the raw columns don't capture

Beyond cleaning and encoding, we derived features that encode churn intuition directly:

eqp_per_month

Handset age relative to how long they've been a customer

usage_declining

Flag for customers whose minutes are trending down

care_intensity

Customer-care calls per minute of use — a frustration proxy

rev_per_min

Revenue efficiency — value of each minute used

drop_rate

Share of attempted calls that fail — network-quality pain

Preprocessing pipeline

Drop ID columns
(no predictive value)

Encode categoricals
(missing kept as its
own level)

Impute numeric gaps
with medians

Engineer 5 domain
features

70/30 stratified split
random_state=42

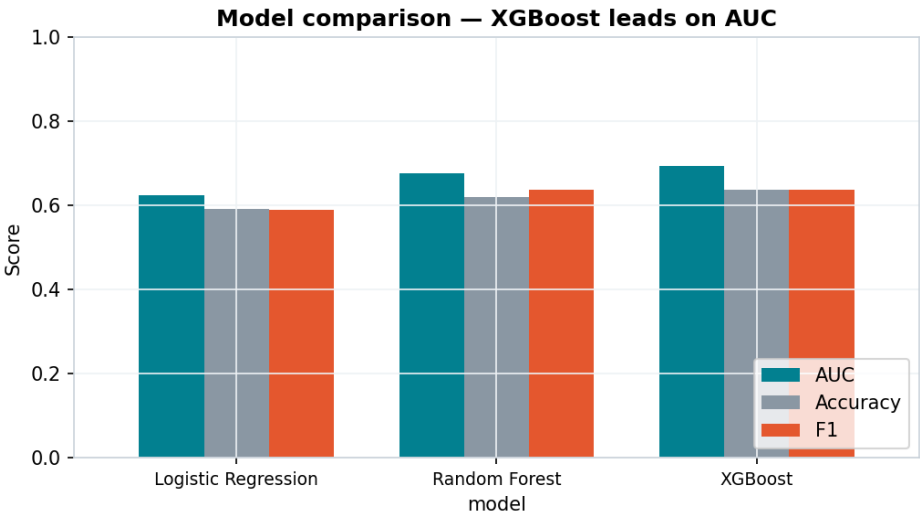
We compared three models — and named the winner

Model	AUC	Accuracy	Precision	Recall	F1
Logistic Regression	0.625	0.591	0.586	0.593	0.590
Random Forest	0.676	0.620	0.604	0.676	0.638
XGBoost	0.695	0.637	0.632	0.642	0.637

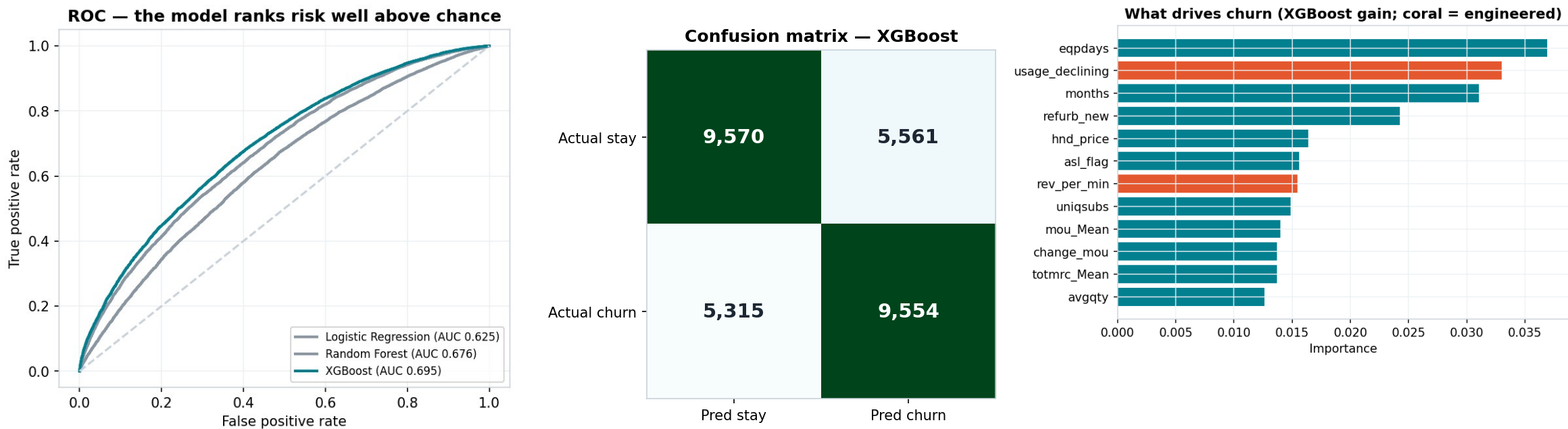
Chosen model: XGBoost

Evaluation metric: ROC-AUC = 0.695. We report AUC (not just accuracy) because it measures how well the model ranks risk regardless of the cut-off the business picks — the property the targeting use-case depends on.

A no-skill model scores 0.50; ours clears that comfortably, so the score reflects real signal.

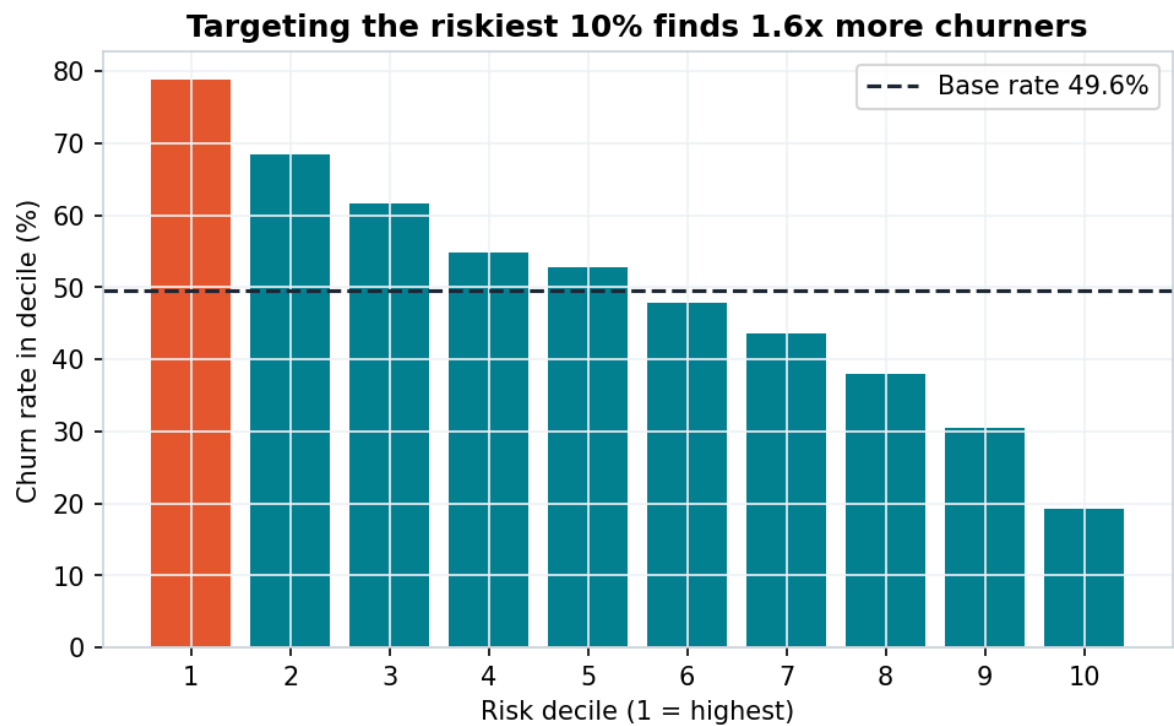


How the model performs and what it keys on



The model ranks well above chance (ROC, left). Top drivers (right) line up with the EDA — equipment age, usage change, and care intensity — so the story is consistent and explainable.

From risk score to action: target the riskiest decile



The core insight

If we rank everyone by predicted risk and contact only the top 10%, that group churns at 79% — about 1.6x the base rate.

Capture

That single decile contains roughly 16% of all churners; the top two deciles capture about 30%.

This is what makes the budget go further — spend retention dollars where churn actually is.

Deciles on the held-out test set; decile 1 = highest predicted risk.

The proposal: a model-driven retention program

1

Score

Rank the full base monthly by churn risk using the model.

2

Target

Take the top-risk decile, prioritising early-tenure, high-value customers.

3

Act

Trigger a tailored save offer — handset upgrade or plan fix — addressing the driver.

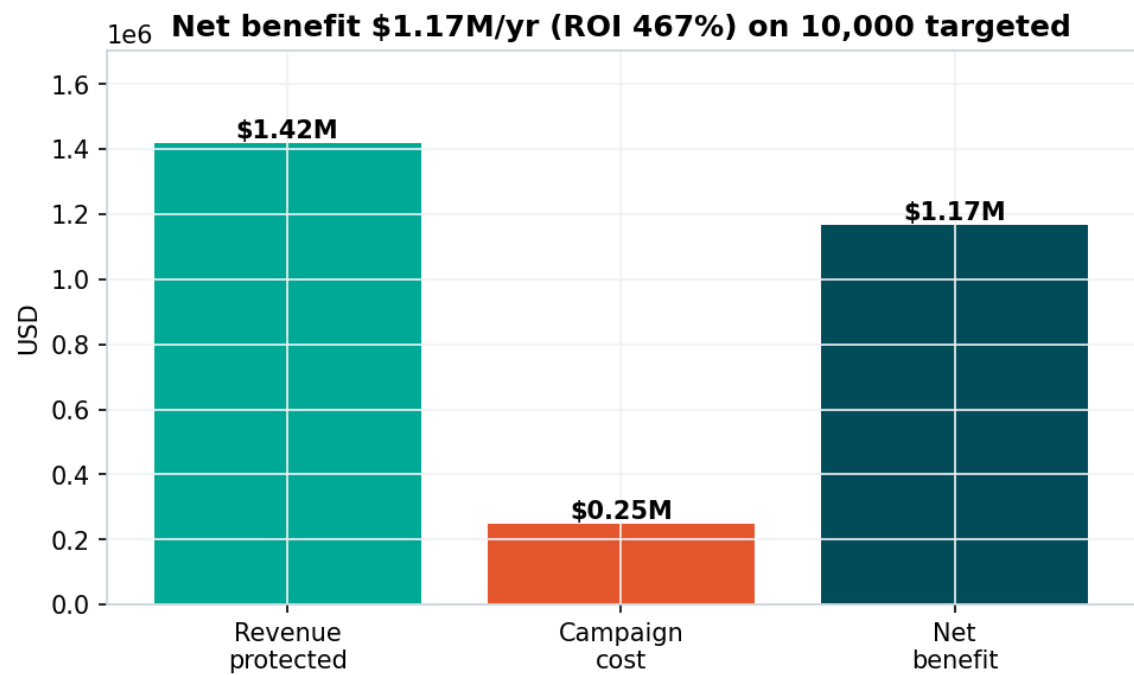
4

Measure

Track save rate vs a hold-out control; feed results back to retrain.

Why it works: the model concentrates spend on customers who would actually leave, and the intervention targets a driver the EDA proved is real (ageing equipment). Each step is measurable.

Quantified impact: the program pays for itself



Worked example (per year, scaled to 100k customers)

- Customers contacted (top 10%)
10,000
- Would-be churners in that group
7,880
- Customers saved (30% offer success)
2,364
- Revenue protected (12 mo × \$50)
\$1.42M
- Campaign cost (\$25 × targeted)
\$0.25M
- Net benefit**
\$1.17M
- Return on retention spend**
467%

Assumptions stated, not hidden: ARPU \$50, 12-mo horizon, \$25 offer cost, 30% save rate. Net benefit stays positive across a 10–50% save-rate range (sensitivity tested).

Pricing and next steps

What we deliver

- Production churn-scoring model + monthly refresh
- Ranked risk list wired to the retention workflow
- A/B hold-out design to prove incremental saves
- Quarterly retrain as new data lands

Commercials

Fixed PoC-to-production fee priced at a small fraction of the \$1.2M/yr net benefit it unlocks, plus a monthly retainer for model upkeep.

Risks & limits

Save-rate is an assumption to validate in the pilot; effects like offer cannibalisation and seasonality are measured via the control group.

References

- CustomerGauge, “Average Churn Rate by Industry — 2025 B2B Benchmarks”: telecom ~22–31% annual churn. customergauge.com/blog/average-churn-rate-by-industry
- Tridens Technology, “Why Telecom Customers Churn and How to Measure It” (2025): telecom churn 20–50% annually; retention vs. acquisition economics. tridenttechnology.com/telecom-churn
- BillingPlatform, “Churn Rates by Industry” (2025): telecom ~31% churn, ~78% retention. billingplatform.com/blog/average-churn-rate-by-industry
- Company A telecom dataset (Client.csv, Record.csv) — provided for this assignment.
- Python libraries: pandas, NumPy, scikit-learn, XGBoost, Matplotlib.
- Generative-AI assistance: Claude (Anthropic), for analysis and slide preparation.